

# The empirical recurrence for OEIS A198447, proved by a two-line transfer matrix

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## Abstract

OEIS A198447 counts arrays in which every cell is constrained by its neighbours on *both* sides, and which are moreover written in canonical form (“values introduced in row major order”). Neither feature is visible to a one-line transfer matrix: the cell condition reaches the row above and the row below, and the canonical-form clause is not local at all. Both are dealt with. What is counted is equality patterns, so  $3!a(n) = \sum_i \binom{3}{i} D_{3-i} L_i(n)$  with  $D_m$  the derangement numbers and  $L_i$  the count over an  $i$ -letter alphabet; and each  $L_i$  is a walk count whose vertices are ordered pairs of consecutive rows. Relabelling-invariance makes the chain strongly lumpable, cutting 98 pairs down to  $S = 23$  states. The entry’s empirical recurrence of order 3, contributed by R. H. Hardin, then follows from a finite exact computation.

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## 1 The sequence and the conjecture

OEIS A198447 is “Number of  $2n \times 2$  arrays with values introduced in row major order and each element unequal to exactly two horizontal and vertical neighbors.”. It has offset 1 and begins

$$3, 13, 71, 433, 2763, 17941, 117263, 768313, \dots$$

The entry carries the following comment:

Empirical:  $a(n) = 10a(n-1) - 25a(n-2) + 16a(n-3)$ .

As of the “Last modified” line on the live entry (May 14 2018 10:30:32, revision 8) the statement is still recorded as empirical, and nothing on the entry records it as settled either way.

Written out, the claim is

$$a(n) = 10a(n-1) - 25a(n-2) + 16a(n-3) \tag{1}$$

for all  $n > 3$ .

## 2 The condition, and what the canonical-form clause counts

The arrays have 2 columns and  $L = 2n + 0$  rows. Write  $x_{t,u}$  for the entry in row  $t$  and column  $u$ , with the neighbour offsets

$$\mathcal{N} = \{(-1, 0), (1, 0), (0, -1), (0, 1)\}$$

written as (row shift, column shift), the requirement is

$$\#\{(d_1, d_2) \in \mathcal{N} : (t+d_1, u+d_2) \text{ lies in the array and its entry is different from } x_{t,u}\} = 2 \tag{2}$$

for every cell  $(t, u)$ .

The condition is stated purely in equalities between entries, so it is unchanged by any relabelling of the values: it is a property of the *equality pattern* — the partition of the cells into classes of equal entries — and not of the values themselves. The clause “values introduced in row major order” selects exactly one array from each relabelling class, so with  $K = 3$  available values the entry counts admissible patterns with at most  $K$  classes.

**Lemma 1.** *Let  $L_i(n)$  count the arrays over an alphabet of  $i$  letters that satisfy the cell condition, with no canonical-form clause, and  $N_j(n)$  the admissible patterns with exactly  $j$  classes. Then  $L_i = \sum_j N_j i(i-1)\cdots(i-j+1)$  and*

$$3! \sum_{j \leq 3} N_j(n) = \sum_{i=0}^3 \binom{3}{i} D_{3-i} L_i(n), \quad D_m = m! \sum_{t=0}^m \frac{(-1)^t}{t!}.$$

*Proof.* A labelled array satisfying the condition is a pattern together with an injection of its classes into the alphabet, and a pattern with  $j$  classes admits  $i(i-1)\cdots(i-j+1)$  of them; that is the first identity. Writing  $i(i-1)\cdots(i-j+1) = j! \binom{i}{j}$  it reads  $L_i = \sum_j (j! N_j) \binom{i}{j}$ , so binomial inversion gives  $j! N_j = \sum_i (-1)^{j-i} \binom{j}{i} L_i$ . Summing over  $j \leq 3$  and exchanging the order of summation, the coefficient of  $L_i$  is  $\sum_{t \leq 3-i} (-1)^t / (i! t!)$ , which multiplied by  $3!$  is  $\binom{3}{i} (3-i)! \sum_{t \leq 3-i} (-1)^t / t! = \binom{3}{i} D_{3-i}$ .  $\square$

### 3 Each $L_i$ is a walk count on pairs of rows

Fix  $i$  and let  $V_i = \{0, \dots, i-1\}^2$ . Because (2) at a cell of row  $t$  involves rows  $t-1$ ,  $t$  and  $t+1$ , a single row is not enough state. Take the ordered pairs  $(p, c) \in V_i \times V_i$  as vertices and declare  $(p, c) \rightarrow (c, x)$  an edge when every cell of the middle row  $c$  satisfies (2) with  $p$  above it and  $x$  below it. Let  $P_i$  be the adjacency matrix,  $\mathbf{s}_i$  the indicator of the pairs  $(p, c)$  whose first row  $p$  satisfies the condition with no row above, and  $\mathbf{e}_i$  the indicator of the pairs whose second row  $c$  satisfies it with no row below.

**Lemma 2.** *For  $L \geq 2$ , the arrays of  $L$  rows satisfying the condition are exactly the walks  $(r^{(1)}, r^{(2)}) \rightarrow (r^{(2)}, r^{(3)}) \rightarrow \dots \rightarrow (r^{(L-1)}, r^{(L)})$  of length  $L-2$  that start at a vertex marked by  $\mathbf{s}_i$  and end at one marked by  $\mathbf{e}_i$ , so  $L_i = \mathbf{s}_i^\top P_i^{L-2} \mathbf{e}_i$ .*

*Proof.* A walk of that form is exactly a sequence of rows  $r^{(1)}, \dots, r^{(L)}$ , since consecutive vertices overlap in one row. In such a walk the condition at the cells of row  $t$  is tested once, at the step leaving  $(r^{(t-1)}, r^{(t)})$ , and with the correct neighbours  $r^{(t-1)}$  and  $r^{(t+1)}$ ; the cells of row 1 are tested by  $\mathbf{s}_i$ , which uses the condition with the row above absent, and those of row  $L$  by  $\mathbf{e}_i$ . So every cell is tested exactly once and with its true neighbourhood, which is the claim.  $\square$

**Lemma 3.** *Write  $\pi(p, c)$  for the equality pattern of the pair. If  $\pi(p, c) = \pi(p', c')$  then for every pattern  $Q$  the two vertices have the same number of successors of pattern  $Q$ , and  $\mathbf{s}_i, \mathbf{e}_i$  take the same value at both. Hence the chain is strongly lumpable over  $\pi$  and  $L_i$  is computed on the quotient, whose states are the set partitions of the  $2 \times 2$  cells of a pair into at most  $i$  classes.*

*Proof.* Equal patterns means  $(p', c') = (\sigma(p), \sigma(c))$  for a permutation  $\sigma$  of the alphabet. The edge condition and the two boundary conditions are statements about equalities between entries, so they are invariant under  $\sigma$ , and  $x \mapsto \sigma(x)$  is a bijection between the successor sets preserving patterns. The number of pairs with a given pattern  $P$  is  $e_P = i(i-1)\cdots(i-|P|+1)$ , the number of injections of its classes into the alphabet, which supplies the weights on the quotient.  $\square$

Assembling the 3 lumped chains block-diagonally into  $N$  of size  $S = 23$ , with the weights of Lemma 1 folded into the start vector  $\mathbf{v}$  and the end-of-array indicator into  $\mathbf{u}$ ,

$$3! a(n) = \mathbf{v}^\top N^{L-2} \mathbf{u}, \quad L = 2n + 0. \quad (3)$$

## 4 The criterion, and the computation

Let  $q(t) = t^3 - \sum_i c_i t^{3-i}$  be the characteristic polynomial of (1) and put  $G = N^2$ , so that consecutive values of  $n$  advance the walk by 2 rows.

**Lemma 4.** *With  $h_j = G^j N^o \mathbf{u}$  for the offset  $o$  that aligns the first admissible index,*

$$3! \left( a(n) - \sum_i c_i a(n-i) \right) = \mathbf{v}^\top G^{j-3} \left( h_3 - \sum_i c_i h_{3-i} \right)$$

*for the corresponding  $j$ . Hence (1) holds from that point on if and only if  $\mathbf{v}^\top G^m w = 0$  for every  $m \geq 0$ , where  $w = h_3 - \sum_i c_i h_{3-i}$ ; and it is enough to check  $m = 0, 1, \dots, S-1$ .*

*Proof.* Substitute (3) and factor. The scalar  $3!$  is nonzero, so it does not affect vanishing. For the last clause, Cayley–Hamilton applied to  $G$  makes  $G^S$  an integer combination of  $I, G, \dots, G^{S-1}$ , so each  $\mathbf{v}^\top G^m w$  with  $m \geq S$  is the same combination of the earlier values; if those vanish, so do all the rest.  $\square$

**Theorem 5.** *The recurrence (1) holds for every  $n > 3$ , which is the statement of Section 1.*

*Proof.* The lumped transition counts were built by taking one representative pair for each pattern, enumerating every row  $x$  over the alphabet, testing (2) on the middle row, and sorting the results by the pattern of the new pair. The vector  $w$  was formed by 3 applications of  $G$  in exact integer arithmetic and  $\mathbf{v}^\top G^m w$  evaluated for  $m = 0, 1, \dots$ , again exactly. Every one of those integers vanishes from the index corresponding to  $n = 3+1$  onwards, and the run was continued until  $S+1$  consecutive zeros had been seen, which by Lemma 4 settles every larger  $m$ .  $\square$

## 5 Verification

Three checks, each independent of the algebra above.

First, the whole construction was compared against the entry’s DATA: the right-hand side of (3), divided by  $3!$ , was evaluated at every one of the 20 published indices and agrees exactly, the quotient coming out an integer each time. This tests the modelling and not only the arithmetic — the inversion of Lemma 1, the reading of (2), the treatment of the first and last rows, and the alignment of the entry’s index with  $L = 2n + 0$  would each show up as a mismatch.

Second, the lumped chain was checked against the unlumped one on the small shapes of this family: the walk counts over the full alphabet agree with  $\sum_P e_P f_m(P)$  term by term.

Third, the recurrence (1) was evaluated on the published terms in exact integer arithmetic with no matrices involved, and the annihilation test of Lemma 4 was carried to the full Cayley–Hamilton bound rather than to a sampled prefix.

## References

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